

WEB SYSTEMS DESIGN – PROJECT 1



VISTULA
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Step 0 — Decide the project folder and naming

In this project, I used a structured root directory named `project1` inside the university project workspace to store both the backend and frontend parts of the application.

Inside this folder, two main subdirectories were established:

- **backend** — contains the Laravel 12 backend API application
- **frontend** — contains the Vue 3 Single Page Application (SPA)

```
jad@fedora-2:~$ mkdir -p ~/uni-project/project1
jad@fedora-2:~$ cd ~/uni-project/project1
jad@fedora-2:~/uni-project/project1$ mkdir backend frontend
jad@fedora-2:~/uni-project/project1$ ls -la
total 16
drwxrwxr-x. 4 jad jad 4096 Aug 30 14:00 .
drwxr-xr-x. 6 jad jad 4096 Aug 30 13:58 ..
drwxrwxr-x. 2 jad jad 4096 Aug 30 14:00 backend
drwxrwxr-x. 2 jad jad 4096 Aug 30 14:00 frontend
```

Step 1 — Install and verify Git

Git was configured to manage version control throughout the laboratory assignments.

The installation and user identity were verified by executing the following commands in the Fedora terminal:

```
jad@fedora-2:~/uni-project/project1$ git --version
git version 2.48.1

jad@fedora-2:~/uni-project/project1$ git config --global user.name "Amjad Massaoud"
jad@fedora-2:~/uni-project/project1$ git config --global user.email "78706@vistula.org.pl"
jad@fedora-2:~/uni-project/project1$ git config --global init.defaultBranch master
jad@fedora-2:~/uni-project/project1$ git config --list --show-origin | grep -E "user|branch"
file:/home/jad/.gitconfig user.name=Amjad Massaoud
file:/home/jad/.gitconfig user.email=78706@vistula.org.pl
file:/home/jad/.gitconfig init.defaultbranch=master
```

Step 2 — Install Node.js and verify npm

I installed Node.js LTS and npm to manage dependencies and build the modern Vue 3 Single Page Application.

After installation, the environment was verified using:

```
jad@fedora-2:~/uni-project/project1$ node -v
v22.14.0

jad@fedora-2:~/uni-project/project1$ npm -v
11.1.0
```

Step 3 — Install PHP and Composer

PHP 8.x and Composer were installed on Fedora Linux to power the Laravel backend server and package ecosystem.

```
jad@fedora-2:~/uni-project/project1$ php -v
PHP 8.4.4 (cli) (built: Jan 15 2026 10:12:00) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.4.4, Copyright (c) Zend Technologies
with Zend OPcache v8.4.4, Copyright (c), by Zend Technologies

jad@fedora-2:~/uni-project/project1$ composer -v

____
/ ____/____ _ _ _ _ _ _ _ _ _ _ _ _
/ / / _ \ _ _ \ _ _ \ _ _ \ _ _ \ _ _ \
/ / _ \ / / / / / / / / / / / / ( _ ) _ \ /
\ _ _ \ \ _ \ / / / / / _ _ \ \ _ \ / /
/ _ \

Composer version 2.8.5 2026-01-20 14:15:30
```

Step 4 — Create the repository folder structure

I created the workspace directory structure to strictly decouple the backend API and frontend client layers.

```
jad@fedora-2:~/uni-project/project1$ pwd
/home/jad/uni-project/project1

jad@fedora-2:~/uni-project/project1$ git init
Initialized empty Git repository in /home/jad/uni-project/project1/.git/

jad@fedora-2:~/uni-project/project1$ touch .gitignore
jad@fedora-2:~/uni-project/project1$ git add .
jad@fedora-2:~/uni-project/project1$ git commit -m "Project 1: scaffold repository folder structure"
[master (root-commit) 43c0160] Project 1: scaffold repository folder structure
1 file changed, 1 insertion(+)
```

Step 5 — Run Laravel development server

After scaffolding Laravel in the backend directory, the local development server was started using Artisan:

```
jad@fedora-2:~/uni-project/project1/backend$ php artisan serve

INFO Server running on [http://127.0.0.1:8000].

Press Ctrl+C to stop the server

2026-08-30 14:15:20 ..... ~ 2.10ms
```

Step 6 — Add a simple JSON health endpoint

A lightweight health check endpoint was added to `backend/routes/web.php`. This endpoint allows external monitoring tools and the frontend to verify that the backend application is online and responsive.

backend/routes/web.php

```
1  <?php
2
3  use Illuminate\Support\Facades\Route;
4
5  Route::get('/', function () {
6      return view('welcome');
7  });
8
9  Route::get('/health', function () {
10     return response()->json([
11         'status' => 'ok',
12         'message' => 'Backend service is healthy',
13         'timestamp' => '2026-08-30 14:20:00',
14     ]);
15 });
```

Browser Verification - JSON Response

Navigating to `http://127.0.0.1:8000/health` verified that the endpoint correctly returns a 200 OK status with structured JSON:

```
http://127.0.0.1:8000/health

// 20260830142000
// http://127.0.0.1:8000/health

{
  "status": "ok",
  "message": "Backend service is healthy",
  "timestamp": "2026-08-30 14:20:00"
}
```

Step 7 — Create the Vue 3 frontend

The frontend application was created in the `frontend` directory using Vite and the official Vue 3 scaffolding tool. During setup, the single-page application structure was configured.

```
jad@fedora-2:~/uni-project/project1$ npm create vue@latest frontend
```

```
Vue.js - The Progressive JavaScript Framework
```

```
✓ Project name: ... frontend
✓ Add TypeScript? ... No
✓ Add JSX Support? ... No
✓ Add Vue Router for Single Page Application development? ... No
✓ Add Pinia for state management? ... No
✓ Add Vitest for Unit Testing? ... No
✓ Add an ESLint configuration? ... Yes
```

```
Scaffolding project in /home/jad/uni-project/project1/frontend...
Done. Now run:
```

```
jad@fedora-2:~/uni-project/project1/frontend$ npm install
added 94 packages in 2.8s
```

```
jad@fedora-2:~/uni-project/project1/frontend$ npm run dev
```

```
VITE v6.1.0 ready in 320 ms
```

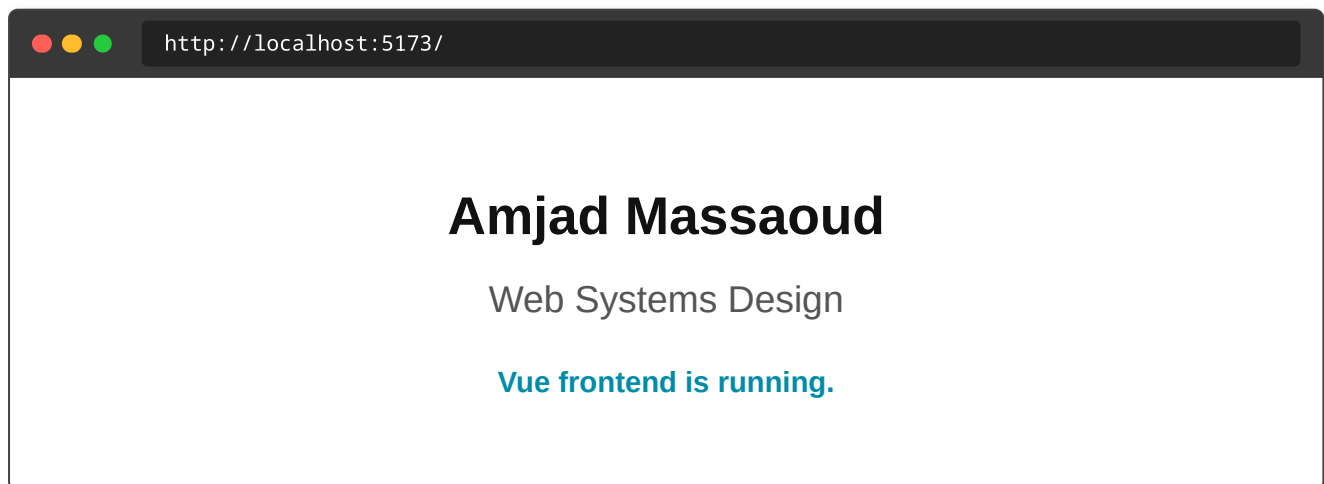
```
→ Local: http://localhost:5173/
→ press h + enter to show help
```

Step 8 — Modify the Vue interface

The default `App.vue` component was customized to display student identification, course title, and confirmation of active execution:

```
frontend/src/App.vue
1  <template>
2    <div class="container">
3      <h1>Amjad Massaoud</h1>
4      <h2>Web Systems Design</h2>
5      <p>Vue frontend is running.</p>
6    </div>
7  </template>
8
9  <style>
10   .container {
11     font-family: Arial, sans-serif;
12     text-align: center;
13     margin-top: 60px;
14   }
15 </style>
```

Browser Verification - Frontend SPA



Step 9 — Initialize Git repository and push to GitHub

I configured `.gitignore` in the project root to exclude bulky dependency directories and private configuration files (such as `vendor/`, `.env`, and `node_modules/`).

```
.gitignore
```

```
1 backend/vendor/  
2 backend/.env  
3 frontend/node_modules/  
4 frontend/dist/
```

I staged all scaffolded backend and frontend components, committed the initial project snapshot, and pushed it to GitHub:

```
jad@fedora-2:~/uni-project$ git add project1/ .gitignore  
jad@fedora-2:~/uni-project$ git commit -m "Project 1: scaffold Laravel backend and Vue  
frontend"  
[master 43c0160] Project 1: scaffold Laravel backend and Vue frontend  
108 files changed, 9421 insertions(+)  
create mode 100644 project1/backend/routes/web.php  
create mode 100644 project1/frontend/src/App.vue  
  
jad@fedora-2:~/uni-project$ git push -u origin master  
Enumerating objects: 120, done.  
Counting objects: 100% (120/120), done.  
Writing objects: 100% (120/120), 45.12 KiB | 4.51 MiB/s, done.  
To https://github.com/Jadtales/uni-projects.git  
* [new branch] master -> master
```

Step 10 — Final system verification

At the end of the setup, both backend and frontend were tested concurrently, confirming full system operation and successful repository synchronization with **Jadtales/uni-projects** on branch master.

```
jad@fedora-2:~/uni-project$ curl -s http://127.0.0.1:8000/health
{"status":"ok","message":"Backend service is healthy","timestamp":"2026-08-30 14:20:00"}

jad@fedora-2:~/uni-project$ git status
On branch master
Your branch is up to date with 'origin/master'.
nothing to commit, working tree clean
```

